

Mock Olympiad (IMC), Round 2

July 19, 2026

Time allowed: 5 hours. Each problem is worth 10 points.

Problem 1. Prove that

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(n+1)} < 2.$$

Problem 2. Compute

$$\lim_{A \rightarrow +\infty} \frac{1}{A} \int_1^A A^{1/x} dx.$$

Problem 3. There are n people in a city, and each of them has exactly 1000 friends (friendship is always symmetric). Prove that it is possible to select a group S of people such that at least $n/2017$ persons in S have exactly two friends in S .

Problem 4. Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a twice differentiable function such that

$$2f'(x) + xf''(x) \geq 1 \quad \text{for } x \in (-1, 1).$$

Prove that

$$\int_{-1}^1 xf(x) dx \geq \frac{1}{3}.$$

Problem 5. Let n be a positive integer, and let $p(x)$ be a polynomial of degree n with integer coefficients. Prove that

$$\max_{0 \leq x \leq 1} |p(x)| > \frac{1}{e^n}.$$